Since electrons in excited states have very small life time, these electrons jump back to ground state through spontaneous emission in a short duration of time (approximately $10^{-8}$ s) by emitting the radiation with same wavelength (or frequency) corresponding to the colours it absorbed (Figure 9.22 (a)). This is called emission spectroscopy.

The wavelengths of these lines can be calculated with great precision. Further, the emitted radiation contains wavelengths both lesser and greater than wavelengths of lines in the visible spectrum.

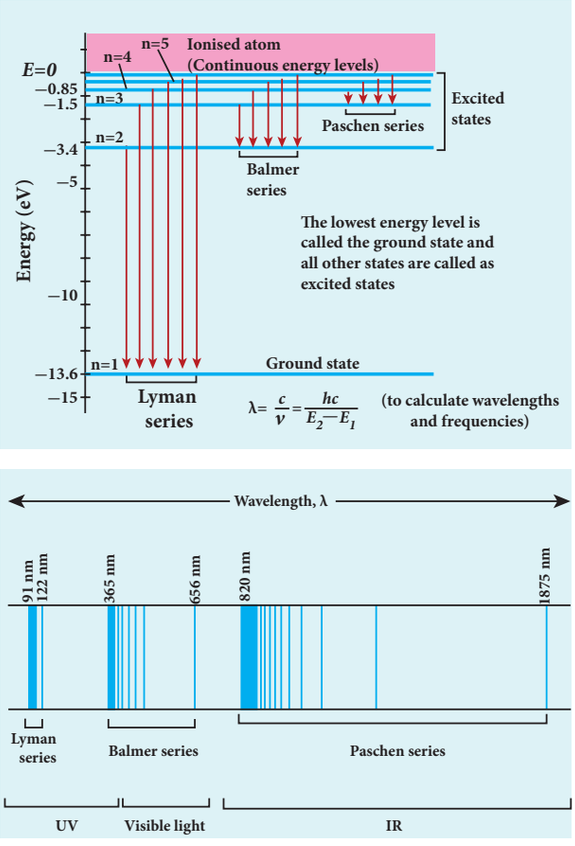



**Figure 9.23** Spectral series – Lyman, Balmer, Paschen series

Notice that the spectral lines of hydrogen as shown in Figure 9.23 are grouped in separate series. In each series, the distance of separation between the consecutive wavelengths decreases from higher wavelength to the lower wavelength, and also wavelength in each series approach a limiting value known as the series limit. These series are named as Lyman series, Balmer series, Paschen series, Brackett series, Pfund series, etc. The wavelengths of these spectral lines perfectly agree with the wavelengths calculate using equation derived from Bohr atom model.

$$\frac{1}{\lambda} = R\left(\frac{1}{n^2} - \frac{1}{m^2}\right) = \bar{\nu} \qquad (9.18)$$

where $\bar{\nu}$ **is known as wave number which is inverse of wavelength,** $R$ is known as Rydberg constant whose value is $1.09737 \times 10^7\ m^{-1}$ and $m$ and $n$ are positive integers such that $m > n$. The various spectral series are discussed below:

### (a) Lyman series

For $n = 1$ and $m = 2,3,4.....$ in equation (9.18), the wave numbers or wavelength of spectral lines of Lyman series which lies in ultra-violet region,

$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{m^2}\right)$$

### (b) Balmer series

For $n = 2$ and $m = 3,4,5.....$ in equation (9.18), the wave numbers or wavelength of spectral lines of Balmer series which lies in visible region,

$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{m^2}\right)$$

### (c) Paschen series

Put $n = 3$ and $m = 4,5,6.....$ in equation (9.18). The wave number or wavelength of spectral lines of Paschen series which lies in infra-red region (near IR) is

$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{3^2} - \frac{1}{m^2}\right)$$

### (d) Brackett series

For $n = 4$ and $m = 5,6,7.....$ in equation (9.18), the wave numbers or wavelength of



XII_Physics_Vol-2_UNIT-9_EM.indd 162 19-12-2022 16:05:05